

B.Tech, 4th SEMESTER
MSE-2 EXAMINATION (BLENDED), EVEN SEM. - 2025-26
AI Driven Full Stack Development (AI308B)

Time: 3 Hours

Total Marks: 15

Note: The project report in PDF format must be uploaded on Moodle during the examination. Part D MSE2_AIFSD

Case Study:

In a college campus, students often lose and find personal belongings. To manage this efficiently, a web-based **Lost & Found Item Management System** is to be developed using the MERN stack.

- Register and log in securely
- Report lost items with relevant details
- Report found items and their location
- View all reported items
- Search items by name or category
- Update or delete their own entries
- Logout securely

Q1	All parts are compulsory	CO-4	
S. N.	Question	BL	Marks
a)	MongoDB Schema Create a MongoDB schema/model for Student with the following fields: User: • Name • Email (Unique) • Password (Hashed) Item: • Item Name • Description • Type (Lost / Found) • Location • Date • Contact Info	3	2
b)	Backend Requirements (Node.js + Express + MongoDB) Auth APIs: • POST /api/register • POST /api/login Item APIs: • POST /api/items → Add item • GET /api/items → View all items • GET /api/items/:id → View item by ID • PUT /api/items/:id → Update item	3	5

	- DELETE /api/items/:id → Delete item - GET /api/items/search?name=xyz → Search Use: bcrypt for password hashing jsonwebtoken (JWT) for authentication		
Q2	All parts are compulsory	CO-3	
S. N.	Question	BL	Marks
a)	Frontend Development (6 Marks) - Create a React application with: - Registration Form (Name, Email, Password) - Login Form (Email, Password) - Dashboard with: - Add item form - Display all items - Search items - Update/Delete items - Logout - Use Axios to connect frontend with backend APIs - On successful login: - Store JWT token in localStorage - Redirect user to Dashboard page Additional Backend Features - Protect route /dashboard using authentication middleware - Ensure only logged-in users can: - View dashboard - Update item details - Handle errors such as: - Invalid login credentials - Duplicate email registration - Unauthorized access - Implement Logout functionality (clear token & redirect) - Add basic UI styling using CSS / Bootstrap	3	5
b)	Deploy this project on GitHub and Render After implementing this project on local machine, deploy it on GitHub and Render. Add the deployment links in the word document.	3	3

- Make Docs and convert it into PDF for submission and it should contain:
 - Code
 - Code output (screenshots)
 - Postman/Thunder client all HTTP request screenshots
 - MongoDB items storage part (screenshots)
 - Render deployment successful (screenshot)
 - Render testing live URL for each end-point

